

Level 8 - supplement

In this build level, new functional blocks are added that help control salient pole PMSMs, such as IPMS, over the entire speed and torque range starting from zero speed. These blocks help to identify the position of rotor while it is in rest or in slow speed by injecting high frequency (HFI) components. At lower speeds where eSMO is not capable of running the motor, HFI can be used and for speeds beyond this range, eSMO can be used. The new functional blocks are given below

- High Frequency Injection (HFI)
- Mode Transition (HFI <-> eSMO)

High Frequency Injection (HFI)

During start up and lower speeds, a voltage of certain magnitude at higher frequency is injected in the D axis. At start up, the rotor position, and hence the D axis location, is unknown. Therefore, the high frequency signal is applied along an assumed D axis. Depending on the location of rotor magnet, or actual D axis, a high frequency current will show up in the assumed Q axis. The magnitude of this Iq current is proportional to the angular error between the assumed and actual D axes. A correction loop is built using this information that will slide the position of assumed D axis into actual D axis, such that high frequency Iq current is zero. A high pass filter is used to extract high frequency signals out of the motor currents, when the motor is spinning.

Initial Position Detection (IPD)

Since the Iq current is proportional to twice the angular error, there is a possibility that the estimated position is 180 deg out of phase at start up. To get out of this tangle, high magnitude voltage pulses are applied on the stator windings to identify the phase current that shows inductance saturation effect. Depending on the polarity of saturating phase current, a window of angle for the position of magnet is identified from which the actual orientation is estimated by adding a 0 or 180 deg to the original estimate.

Mode Transition

This module helps in transitioning the control from HFI to eSMO. At lower speeds, HFI is used and at higher speeds, eSMO is used. However, there is a short range where both methods are used and the control is in transition. It does a balancing act where the D axis angle estimates from HFI and eSMO methods are compared, and weighted based on the running speed as against their limits where the control is either fully HFI or eSMO.